



USER MANUAL

UPS FOR ELEVATOR

BATTERY POWERED EMERGENCY

SOJI SJ-S / 60VDC



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I. GENERAL:

This device is mainly an UPS supplied by a 60V 7AH batteries, with some additional functions.

This device can be used for emergency evacuation operation for all elevator panels with inverters of all power ratings, because the inverters DC buses are directly derived from the batteries.

The dimensions are 155 mm width, 180mm height, and 175mm depth, weight 6.0 kg.

The operating ambient temperature is 0°C – 45°C.

II. SPECIFICATIONS:

The device has the following additional features in addition to a standard UPS:

- In the evacuation state, the device stops after a preselected time, and becomes completely dead and does not consume any current from the batteries. This saves the batteries from discharging during the waiting time for the AC to come back. When AC power comes back, it starts and goes into stand by state.
- In addition to the main uninterrupted output OUTL, there is a time delayed output K1.
- The delay time may be selected by J22 and J23 jumpers, J22:0 position 5 sec, 1 position 10 second, J23:1 position 20s.
- There is a W automat circuit breaker on battery circuit for protection of the batteries and for easy service.
- The device is fully protected against reverse polarity battery connection.
- There is a W automat circuit breaker on 220V AC line, for protection and easy service.

The device functions without battery connection that means the voltage on L1 input is connected to the OUTL output, even if the batteries are not connected, or discharged completely to zero.

So the batteries do not cause any additional fault to the operation of the elevator panel. This improves the reliability of the elevator panel equipped with an emergency evacuation device. Of course, in this case, the emergency evacuation operation cannot be done.



For most of the possible faults of the device, the L1 and OUTL terminals stays connected, and the elevator panel operates. So generally the device does not cause any potential fault to the operation of the elevator panel.

ATTENTION! A standard UPS does not satisfy item 6 and 7 above. A standard UPS with the batteries disconnected, does not start when it is connected to 220V AC for the first start. That means the 220V AC input voltage stays disconnected from the output terminal, and the elevator panel does not function.

III. OPERATION:

1. At the beginning, when L1 phase is OFF, the device is dead, and the contacts are as shown in the figure.
2. When L1 phase becomes ON, the device starts operation, the relays GR and GRX becomes ON, OUTL output is connected to L1 input voltage.
3. When L1 phase fails, GR relay becomes OFF, and if the battery is connected, OUTL continuous to supply the load without any interruption, and the operation duration timing starts. Operation duration is 300 second.

If the battery is not connected, when AC fails, the OUTL fails also, and the devices dies. When L1 phase comes back, the device starts from the beginning again.

4. After 5 or 10 or 20 seconds (can be selected by J22 and J23 jumper) K1 relay becomes ON, for the delayed output.
5. After 300 seconds, the device stops and becomes complete dead. From now on no current is drawn from the batteries, batteries are saved from discharging.
6. When L1 phase comes back, the device starts operation from the beginning.

IV. BUZZER and LEDS:

A. When AC power is ON:

1. MAIN, GR, GRX, CH, 12V, 5V led are ON.



2. When the battery is charging, the BAT led is ON, when completely charged it becomes OFF. It starts bright and dimes gradually from the start to the end of the charging.

3. In case of overload, the buzzer beeps continuously.

B. When AC is OFF, operation from the batteries:

1. MAIN and GR led are OFF, INV led (green), CH, GRX, 12V and 5V led are ON, 5 (or 10, or 20) sec seconds after, K1 led is ON.

2. Buzzer starts intermittent beeping.

If overload occurs, buzzer beeps continuously.

If overload lasts for less than 3 seconds, buzzer starts intermittent beeping again. If overload lasts for more than 3 seconds, the device stops and becomes dead.

3. With normal load, when the battery voltage drops below 56.0V, buzzer beeps with higher rate than normal intermittent rate. When the voltage drops below 49.0V, the device stops and dies.

V. WARNINGS, DIAGNOSES and CORRECTIONS:

The batteries must be in good condition for a problem free operation. For this:

1. Measure the terminal voltages of each battery, they should be around 12-13 Volts. During this measurement the batteries must be loaded by a few amperes.

2. Measure the terminal voltages of each battery during charging, do not use the batteries lower than 11.0 Volt terminal voltages after 12 hours.

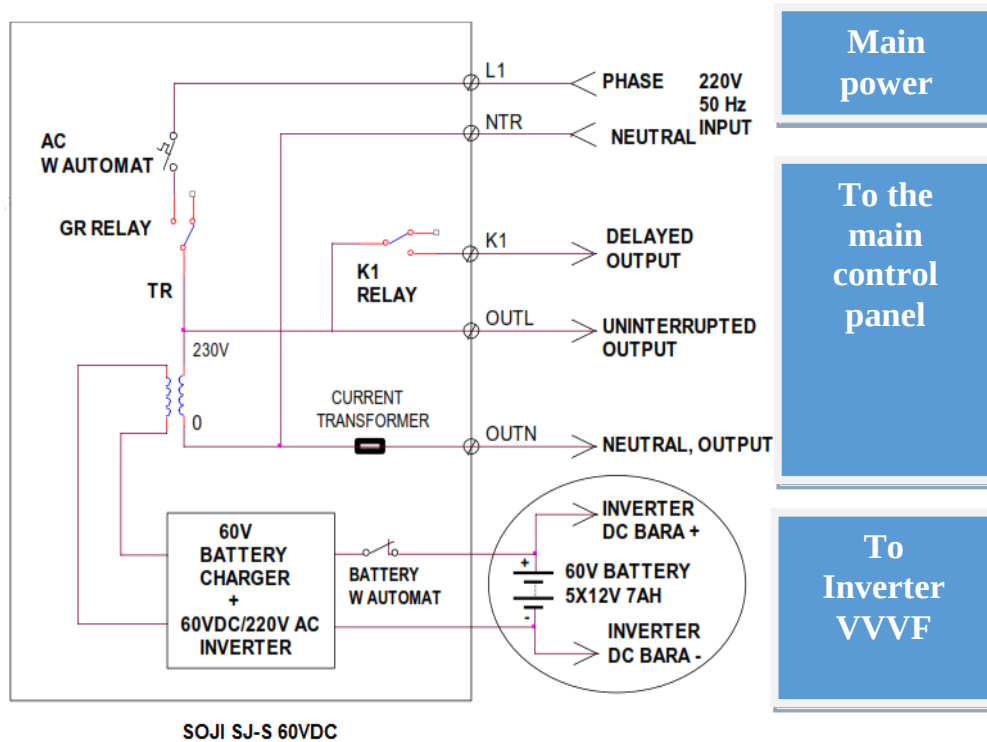
3. When the total terminal voltage is around 68-69 Volts after the charging, and if it drops below 45V or lower at the beginning of the first evacuation operation, keep them for charging at least 20 hours. If they drops again below 45 Volt at the beginning of the operation, replace them by new batteries.



VI. PERIODICAL MAINTENANCE:

1. For a long service life of the batteries the ambient temperature should be limited to maximum 50C degree. Since the maximum temperature in the machine room is limited to 45C degree, this condition may be satisfied.
2. Read item 6 for the maintenance of the batteries. Additionally, a periodical discharge and charge of the batteries helps to activate the batteries for their maximum use. To do this, let the device make an evacuation operation at least once during the periodical maintenance work of the elevator once in a month.
3. Keep the device in a clean environment.

VII. BLOCK DIAGRAM



In this connection, the SOJI SJ-S 60VDC is supplied by the 60V battery as in our standard case.

The additional/optional 36V battery is connected in series with the 60V battery.

This 36V additional battery will be charged by the optional 36V charger fixed within the device.

The total 96 V battery will be used to supply the inverter DC bara.

Note: Without connecting the additional 36V battery, this device may be used as our standard device.

ATTENTION! BE CAREFUL NOT CONNECT THE BATTERIES IN REVERSE POLARITY!



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